

Plate Loading Challenge

Problem ID: problem-36-24527669

The University Gym is running a plate-loading challenge. Before each attempt, the instructor present announces a target weight. Your goal is to load the heaviest barbell that does not exceed the announced target. The equipment officer also wants to know how many different selections of plate pairs could produce that desired weight.

The empty bar alone weighs B units. The rack holds n labelled pairs of plates, numbered 1 to n . Pair i contains two plates, each weighing p_i units. To use a pair, load each end of the bar with the plates, adding $2p_i$ units in total. Each pair may only be used at most once, and it cannot be split up or combined with another. There is also enough room on the bar for any number of selections.

2 selections are different if they use different sets of pair labels. Pairs with equal weights still have different labels. Rearranging the selected plates does not create a different selection. Selecting no pairs is allowed and it also counts as 1 selection.

All weights are expressed in integer units, where one unit is = 100 grams. The target T includes the weight of the bar. Find the maximum achievable total weight at most T , and the exact number of selections achieving it.

Input

The first line contains three integers n , B , and T , where $1 \leq n \leq 200$, $1 \leq B \leq 1000$, and $B \leq T \leq B+10000$.

The second line contains n integers $p_1, \dots, p_n$, where $1 \leq p_i \leq 1000$. The i th integer is the weight of *one plate* from pair i , in the same units as B and T . There is one challenge per input file.

Output

Output two integers on one line, separated by a space: the maximum total barbell weight in the given units, and the exact number of selections achieving that weight. The count may exceed the range of a 64-bit integer.

Sample explanations

In Sample 1, the empty bar weighs 200 units (20 kg). The selections $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3, 4\}$ each add 160 units. They are the three selections achieving the target of 360 units.

In Sample 2, pair 1 alone gives 212 units and pair 2 alone gives 220 units. Both pairs together would exceed the target of 221 units, so the answer is 220 units with one selection.

In Sample 3, the target equals the empty bar's weight. The only possible selection is the empty selection.

Sample Input 1

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4 200 360
50 30 30 20
```

Sample Output 1

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360 3
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Sample Input 2

```
2 200 221
6 10
```

Sample Output 2

```
220 1
```

Sample Input 3

```
3 200 200
10 20 30
```

Sample Output 3

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200 1
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